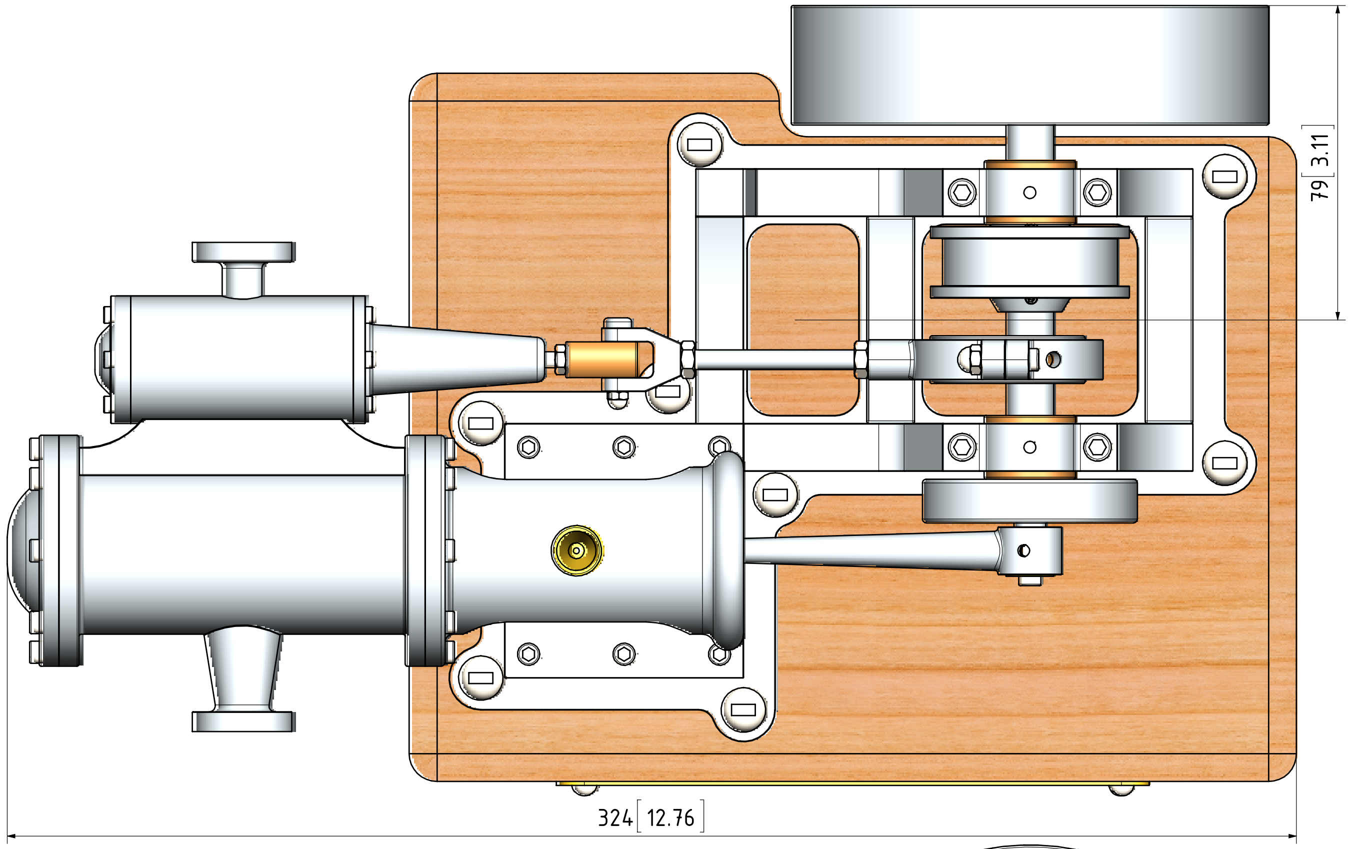

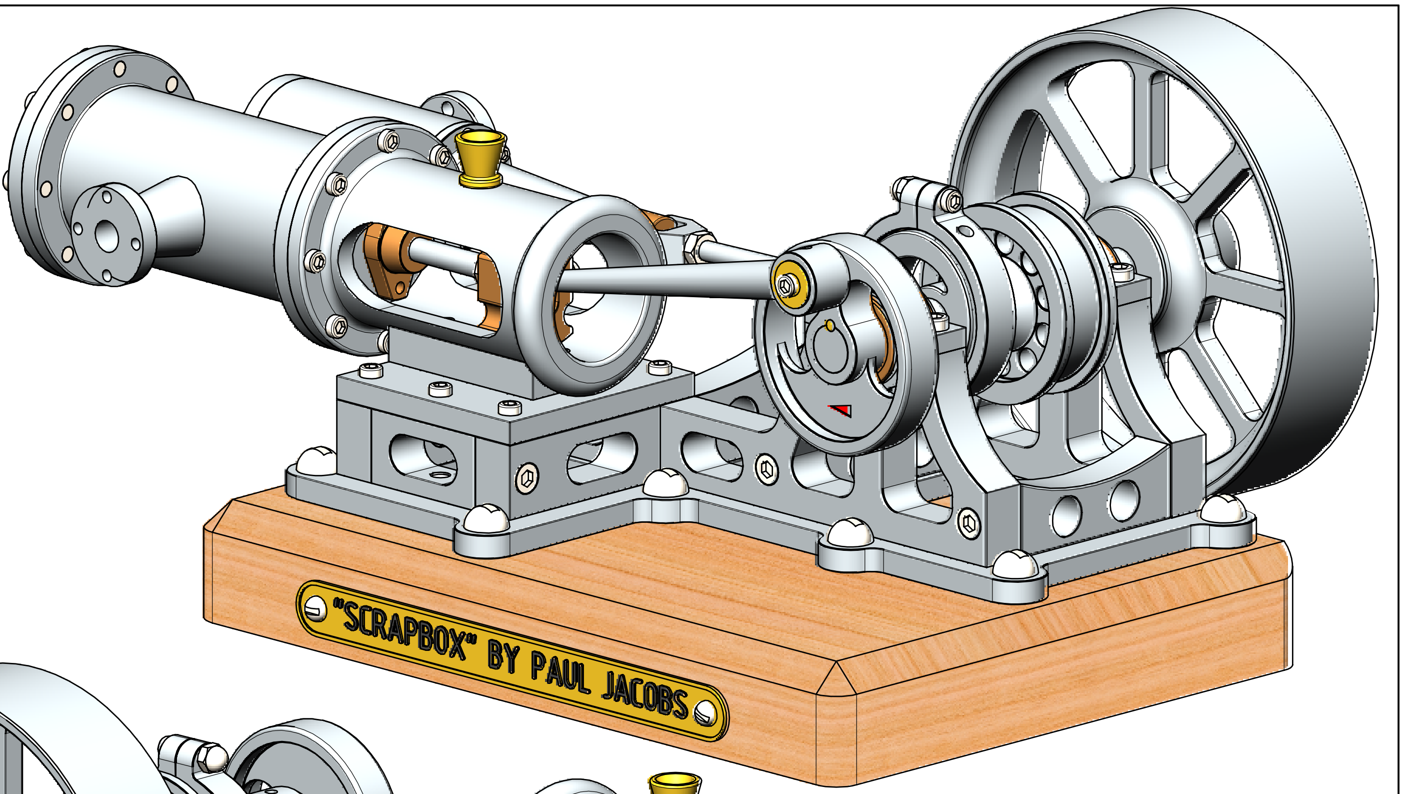

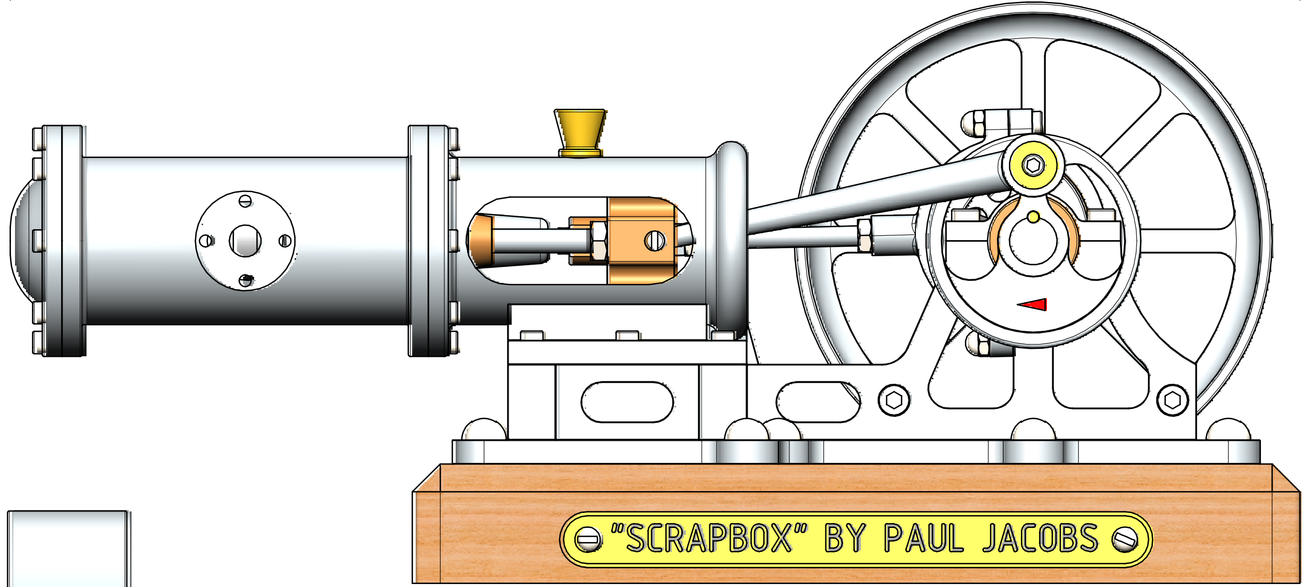

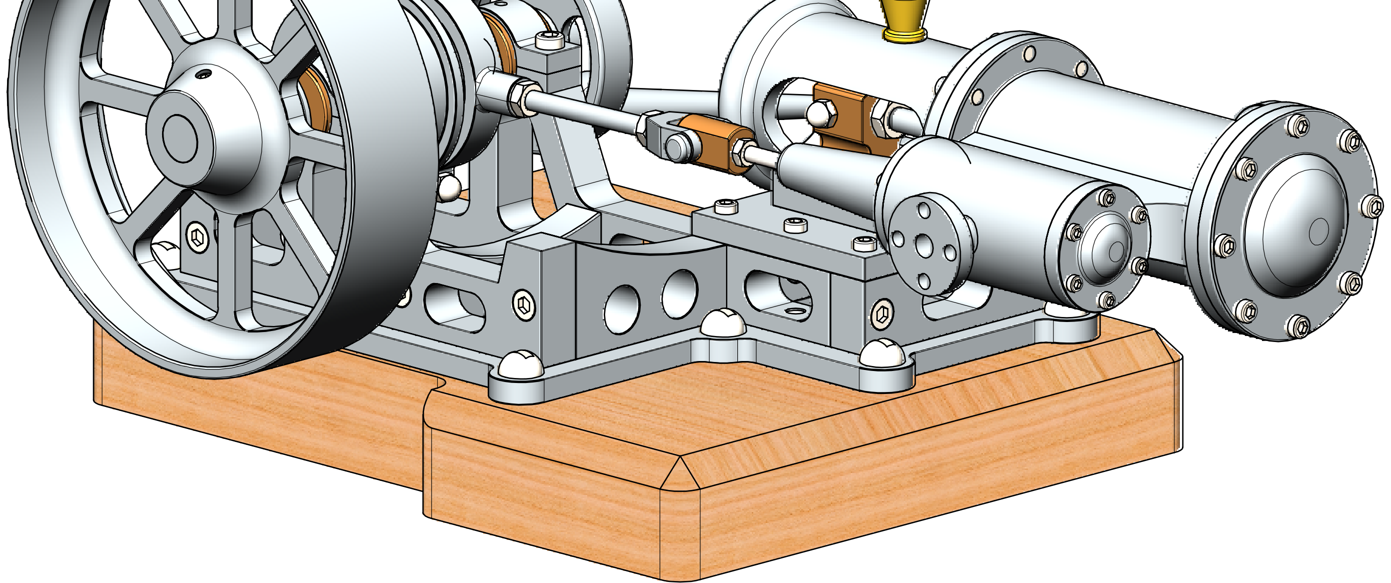

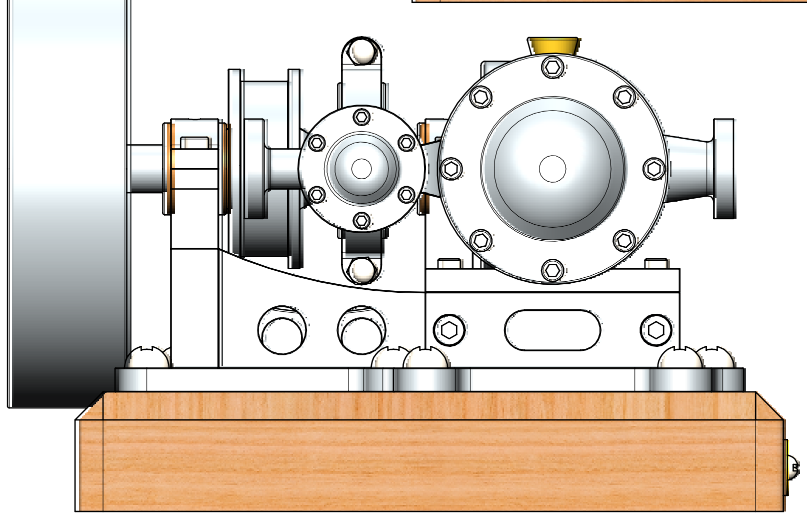

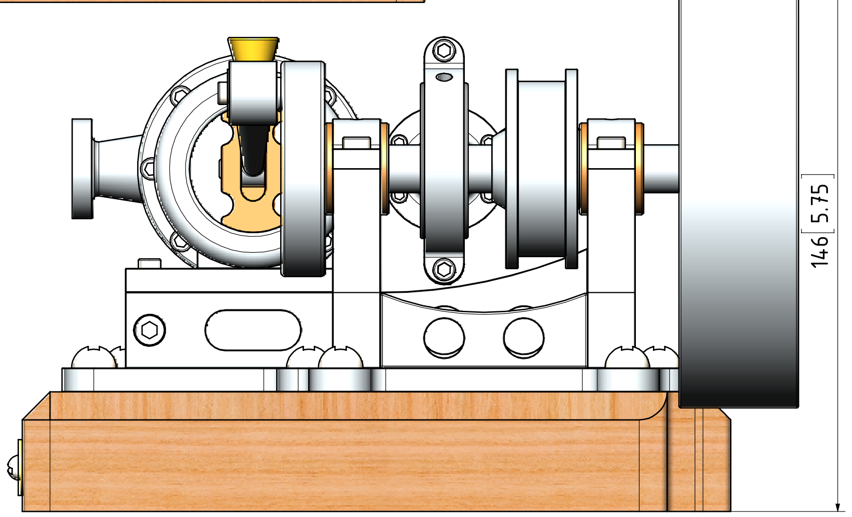

QTY.	PART NUMBER
1	SCRAPBOX-2-05-ECCENTRIC SHEAVE
1	SCRAPBOX-2-06-ECCNTRIC STRAP
1	SCRAPBOX-2-07-FLYWHEEL
1	SCRAPBOX-2-08-DRIVE PULLEY
1	SCRAPBOX-M3 DOME NUT
3	SCRAPBOX-M3x4 A-K GRUB SCREW
12	SCRAPBOX-M3x10 A-K CYL HEAD SCREW
1	SCRAPBOX-M3x15 A-K GRUB SCREW
3	SCRAPBOX-M4 DOME NUT
1	SCRAPBOX-M4 NUT
2	SCRAPBOX-M4x8 A-K GRUB SCREW
23	SCRAPBOX-M4x10 A-K CYL HEAD SCREW
1	SCRAPBOX-M4x12-OD SPECIAL WASHER
3	SCRAPBOX-M4x14 A-K C-SINK SCREW
2	SCRAPBOX-M4x15 A-K CYL HEAD SCREW
2	SCRAPBOX-M4x20-ROUND HEAD WOOD SCREW
2	SCRAPBOX-M5 NUT
2	SCRAPBOX-M5x10 A-K C-SINK SCREW0
4	SCRAPBOX-M5x12 A-K CYL HEAD SCREW
14	SCRAPBOX-M5x14 A-K C-SINK SCREW
9	SCRAPBOX-M5x18 A-K C-SINK SCREW
1	SCRAPBOX-M6 NUT
9	SCRAPBOX-M7x30-ROUND HEAD WOOD SCREW

QTY.	PART NUMBER
1	SCRAPBOX-1-01-BASE FRAME
1	SCRAPBOX-1-02-LH FRAME PLATE
1	SCRAPBOX-1-03-RH FRAME PLATE
2	SCRAPBOX-1-04-REAR CROSS MEMBER
1	SCRAPBOX-1-05-TRUNK GUIDE PEDESTAL
1	SCRAPBOX-1-06-TRUNK GUIDE BASE PLATE
1	SCRAPBOX-1-07-TRUNK GUIDE
1	SCRAPBOX-1-08-PISTON GLAND
1	SCRAPBOX-1-09-CYLINDER
1	SCRAPBOX-1-10-CYLINDER EXHAUST FLANGE
1	SCRAPBOX-1-11-PORT BLOCK
1	SCRAPBOX-1-12-VALVE CYLINDER
1	SCRAPBOX-1-13-WOODEN STAND
1	SCRAPBOX-1-14-NAMEPLATE
1	SCRAPBOX-1-15-TRUNK GUIDE OILER
1	SCRAPBOX-2-01-CRANK DISC+SHAFT
1	SCRAPBOX-2-02-PISTON+ROD+CROSSHEAD
1	SCRAPBOX-2-03-CON-ROD
1	SCRAPBOX-2-04-PISTON VALVE+SPINDLE

NOTES: THIS STEAM ENGINE IS BASED ON DRAWINGS GIVEN TO ME. THE AUTHOR OF THE ORIGINAL DESIGN/DRAWINGS IS PAUL JACOBS. DATE UNKOWN. UNKNOWN MAGAZINE PAGES MARKED 66 TO 79

TITLE
1 CYLINDER HORIZONTAL MILL STEAM ENGINE
WITH CYLINDER CENTER EXHAUST

DRAWING CONTENTS
GENERAL ARRANGEMENT, ISOMETRIC
VIEWS & BILL OF MATRIALS

PROJECT No 09B-18-00
JDW DRAUGHTING SERVICES

J.A.M. DE WAAL. 12 BRIGHTWELL STREET PAPAKURA 2110.
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E-MAIL: dewaal@xtra.co.nz.

PROJECTION
JDWDS
DATE DECEMBER-2017
SHEET: 01 OF 04

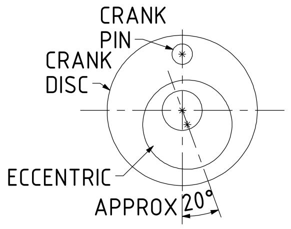
MODEL SCALE: 1:1
DWG SCALE: 1:1 @A3 OR AS SHOWN

A3 No: SCRAPBOX-01

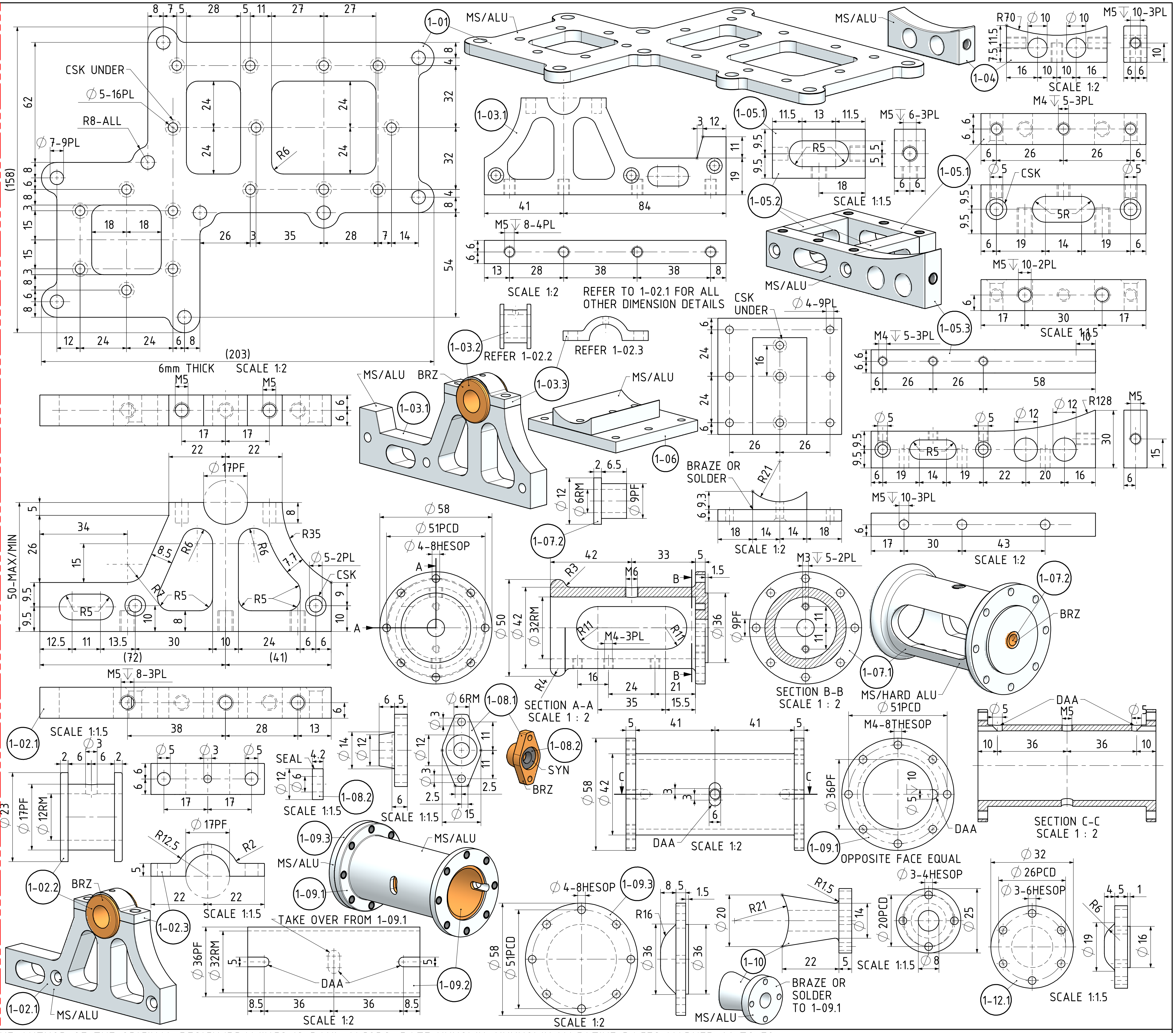
NOTES:
0. ALL DRAWINGS ARE IN METRIC MEASUREMENTS
1. ALL ENGINEERING PRACTICES SHALL BE APPLIED WITH REGARDS TO HOLE AND SHAFT TOLERANCES.
2. WHERE SCREWS OR BOLTS ARE USED THE CLEARANCE HOLES SHALL BE APPROXIMATELY 5% TO 8% LARGER THAN THE MATCHING TAPPED HOLE.
3. PREFERABLY ALL TAPPED HOLES AND MATCHING SCREWS AND/OR BOLTS TO BE METRIC FINE (MF)
4. MATERIALS SPECIFIED ON THE DRAWINGS ARE INDICATIVE ONLY. THE BUILDER CAN MAKE HIS/HER OWN MATERIAL CHOICE.
5. ALL CONNECTIONS/JOINTS WHICH HAVE STEAM PRESSURE APPLIED TO IT SHALL BE SILVER/HARD SOLDERED.
6. COMPRESSION SPRINGS ARE DRAWN IN COMPRESSED STATE (CP), UNCOMPRESSED STATE IS APPROX 40% TO 60% LONGER THEN COMPRESSED STATE.
7. WHERE PREFERRED SCREW OR RIVETED CONNECTIONS CAN BE OMITTED AND PARTS CAN BE BONDED TOGETHER BY USING EITHER HIGH STRENGTH GLUE, EPOXY RESIN, OR SOLDER.
8. PARTS WHICH ARE DIRECTLY EXPOSED TO STEAM AND/OR WATER SHOULD BE CONSTRUCTED USING NON-FERROUS OR NON CORROSIVE MATERIAL SUCH AS BRASS, BRONZE, GUNMETAL, STAINLESS STEEL, COPPER OR MONEL.
9. THE ORDER IN WHICH THE PARTS/COMPONENTS ARE MANUFACTURED AND THE MODEL IS ASSEMBLED IS ENTIRELY LEFT UP TO THE BUILDER/MODEL MAKER.
10. A COLOUR SCHEME FOR THIS PROJECT IS ENTIRELY LEFT UP TO THE MODEL MAKER.
11. THE MANNER IN WHICH THE PARTS/COMPONENTS ARE MANUFACTURED IS ENTIRELY LEFT UP TO THE BUILDER.
12. USE LOCTITE, ON SCREW OR PRESS FIT CONNECTIONS OR SURFACES, WERE DEEMED NECESSARY TO PREVENT PARTS FROM LOOSENING.
13. WASHER SHALL BE USED WHERE DEEMED NECESSARY.
XX. ERRORS AND/OR OMISSIONS MAY OCCUR IN THE DRAWINGS, DO NOT HESITATE TO CONTACT ME SO THAT THE ERRORS/OMISSIONS CAN BE RECTIFIED.

MATERIAL ABBREVIATIONS:
ALU = ALUMINIUM
BRS = BRASS
BRZ = BRONZE OR GUNMETAL (BRZ/GM)
CI = CAST IRON
CU = COPPER
GRA = GRAPHITE
MS = MILD STEEL/BRIGHT MILD STEEL
S/S = SILVER STEEL OR STAINLESS STEEL
SPS = SPRING STEEL
PEEK= POLYETHER ETHER KETONE
SYN = SYNTHETIC MATERIAL SUCH AS VETON, NYLON, TEFLON OR RUBBER IN GENERAL SYNTHETIC MATERIALS SOULD BE ABLE TO WITHSTAND THE HEAT AND PRESSURE(S) APPLIED TO THEM.
nnn/nnn MEANS THAT EITHER MATERIAL CAN BE USED

OTHER ABBREVIATIONS
DP = DEEP
DAA= DRILL AFTER ASSEMBLY
D&TAA= DRILL AND TAP AFTER ASSEMBLY
CF = CLOSE FIT (SIZE FOR SIZE)
PF = PRESS FIT
PFAA= PRESS FIT AFTER ASSEMBLY
PCD = PITCH CIRCLE DIAMETER
RM = REAM
HEX = HEXACON, 6SIDED
CP = COMPRESSED
KNL = KNURLED
CSK = COUNTERSINK
PL = PLACES
DWL= DOWEL
(T)HESOP=(TAPPED)HOLES EQUALLY SPACED ON PCD
(T)HESOC=(TAPPED)HOLES EQUALLY SPACED ON CIRCUMFERENCE
SA-xxx = SUB ASSEMBLY-xxx


CRANK PIN
CRANK DISC
ECCENTRIC
APPROX 20°

THE OFF SET ANGLE OF THE ECCENTRIC IN RELATION TO THE CRANK AXIS TO BE EXPERIMENTALLY DETERMINED FOR THE SMOOTH RUNNING OF THE ENGINE AND SATISFACTION OF THE BUILDER


1-01
MS/ALU
CSK UNDER
Ø5-16PL
R8-ALL
Ø7-9PL
R6
(158)
(203)
6mm THICK
SCALE 1:2
M5
1-02.1
1-02.2
1-02.3
1-03.1
1-03.2
1-03.3
REFER 1-02.2
REFER 1-02.3
REFER TO 1-02.1 FOR ALL OTHER DIMENSION DETAILS
M5 ⌵ 8-4PL
Ø17PF
50-MAX/MIN
Ø5-2PL
CSK
(72)
(41)
M5 ⌵ 8-3PL
SCALE 1:1.5
BRZ
1-04
R70
M5 ⌵ 10-3PL
SCALE 1:2
1-05.1
1-05.2
1-05.3
M5 ⌵ 6-3PL
M4 ⌵ 5-3PL
M5 ⌵ 10-2PL
M5 ⌵ 10-3PL
R128
1-06
CSK UNDER
Ø4-9PL
BRAZE OR SOLDER
R21
1-07.1
1-07.2
Ø58
Ø51PCD
Ø4-8HESOP
M6
M4-3PL
SECTION A-A
SCALE 1 : 2
SECTION B-B
SCALE 1 : 2
M3 ⌵ 5-2PL
MS/HARD ALU
Ø51PCD
M4-8THESOP
1-08.1
1-08.2
SEAL
SYN
SCALE 1:1.5
1-09.1
1-09.2
1-09.3
TAKE OVER FROM 1-09.1
DAA
OPPOSITE FACE EQUAL
SECTION C-C
SCALE 1 : 2
1-10
BRAZE OR SOLDER TO 1-09.1
Ø3-4HESOP
Ø32
Ø26PCD
Ø3-6HESOP
1-12.1

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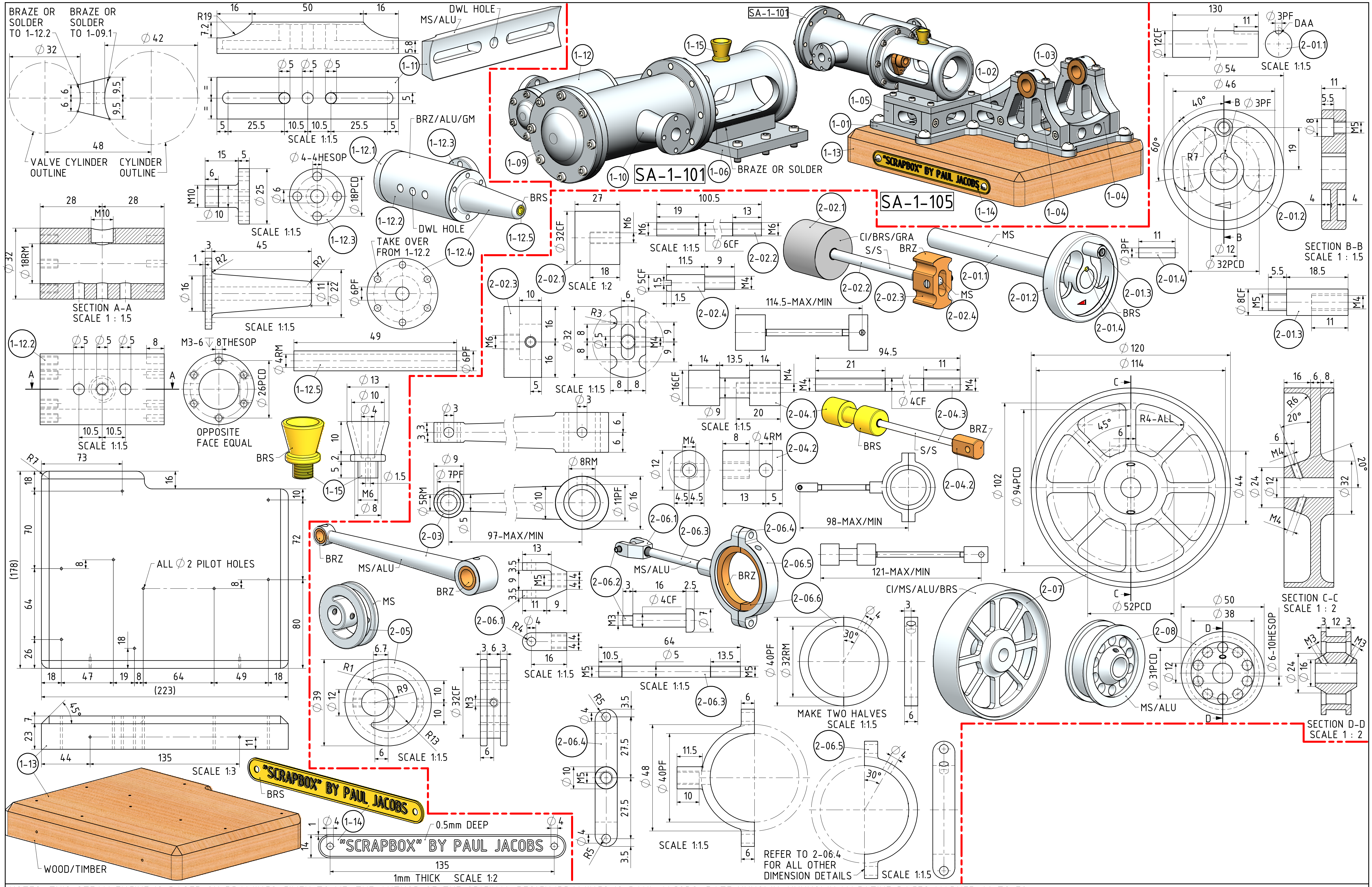
TITLE
1 CYLINDER HORIZONTAL MILL STEAM ENGINE WITH CYLINDER CENTER EXHAUST

DRAWING CONTENTS
NOTES, PARTS AND ASSEMBLIES

PROJECT No 09B-18-00
JDW DRAUGHTING SERVICES
J.A.M. DE WAAL. 12 BRIGHTWELL STREET PAPAKURA 2110.
NEW ZEALAND. PHONE: 0064 09 2988815. MOB: 0211791000
E-MAIL: dewaal@xtra.co.nz
PROJECTION
JDWDS
DATE DECEMBER-2017
MODEL SCALE: 1:1
DWG SCALE: 1:1 @A3 OR AS SHOWN

SHEET: 02 OF 04
A3 No: SCRAPBOX-02


BRAZE OR SOLDER TO 1-12.2
BRAZE OR SOLDER TO 1-09.1
Ø 32
Ø 42
VALVE CYLINDER OUTLINE
CYLINDER OUTLINE
48
DWL HOLE
MS/ALU
1-11
SCALE 1:1.5
BRZ/ALU/GM
1-12.1
1-12.2
1-12.3
1-12.4
1-12.5
4-4HESOP
Ø 18PCD
DWL HOLE
TAKE OVER FROM 1-12.2
BRS
SECTION A-A
SCALE 1 : 1.5
M3-6 ↓ 8THESOP
Ø 26PCD
OPPOSITE FACE EQUAL
1-15
SA-1-101
1-01
1-02
1-03
1-04
1-05
1-06
1-09
1-10
1-12
1-13
1-14
1-15
BRAZE OR SOLDER
SA-1-105
"SCRAPBOX" BY PAUL JACOBS
2-01.1
2-01.2
2-01.3
2-01.4
2-02.1
2-02.2
2-02.3
2-02.4
2-03
2-04.1
2-04.2
2-04.3
2-05
2-06.1
2-06.2
2-06.3
2-06.4
2-06.5
2-06.6
2-07
2-08
CI/BRS/GRA
S/S
BRZ
MS
BRS
SCALE 1:2
SECTION B-B
SCALE 1 : 1.5
114.5-MAX/MIN
98-MAX/MIN
97-MAX/MIN
121-MAX/MIN
CI/MS/ALU/BRS
R4-ALL
SECTION C-C
SCALE 1 : 2
SECTION D-D
SCALE 1 : 2
6-10HESOP
MS/ALU
MAKE TWO HALVES
SCALE 1:1.5
REFER TO 2-06.4 FOR ALL OTHER DIMENSION DETAILS
ALL Ø 2 PILOT HOLES
(178)
(223)
SCALE 1:3
WOOD/TIMBER
"SCRAPBOX" BY PAUL JACOBS
0.5mm DEEP
135
1mm THICK
SCALE 1:2

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TITLE
1 CYLINDER HORIZONTAL MILL STEAM ENGINE
WITH CYLINDER CENTER EXHAUST

DRAWING CONTENTS
NOTES, PARTS AND ASSEMBLIES

PROJECT No 09B-18-00
JDW DRAUGHTING SERVICES

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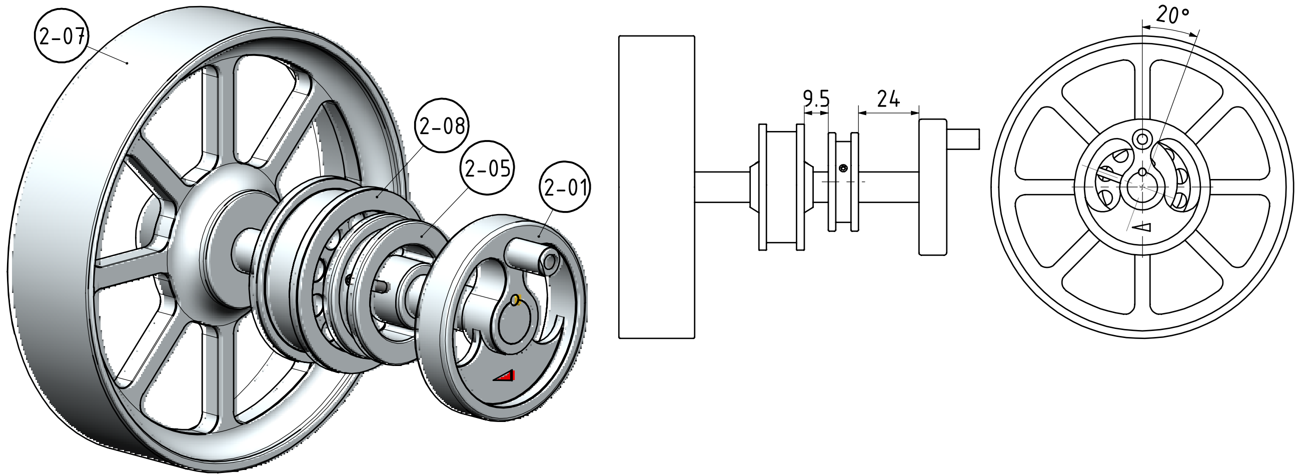
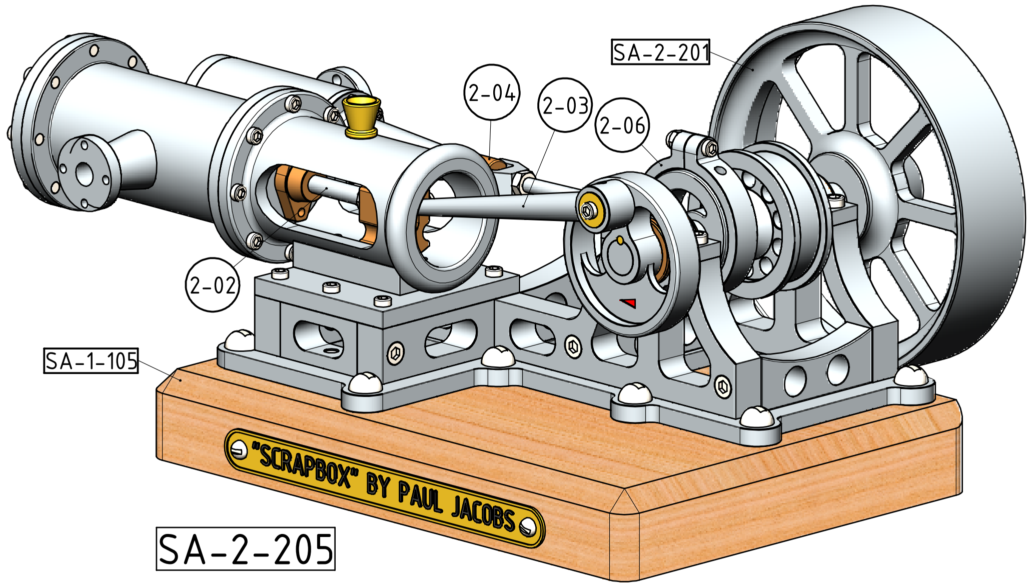
PROJECTION
JDWDS
DATE DECEMBER-2017
MODEL SCALE: 1:1
DWG SCALE: 1:1 @A3 OR AS SHOWN

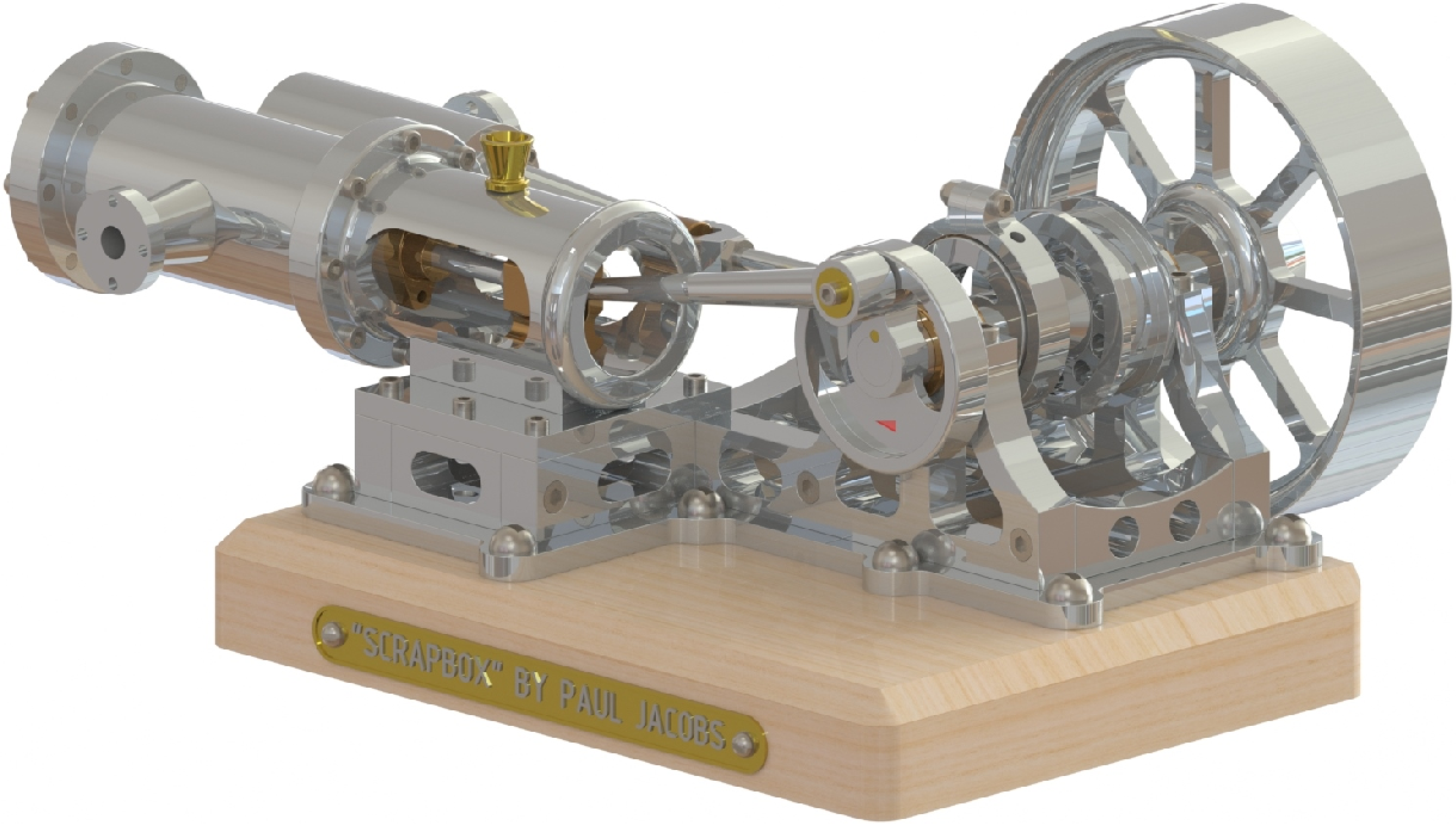
SHEET: 03 OF 04
A3 No: SCRAPBOX-03


2-07
2-08
2-05
2-01
9.5
24
20°

SA-2-201


SA-2-201
2-04
2-03
2-06
2-02
SA-1-105
"SCRAPBOX" BY PAUL JACOBS

SA-2-205


"SCRAPBOX" BY PAUL JACOBS

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TITLE
1 CYLINDER HORIZONTAL MILL STEAM ENGINE
WITH CYLINDER CENTER EXHAUST

DRAWING CONTENTS
ASSEMBLIES AND
RENDERED PICTURES

PROJECT No 09B-18-00
JDW DRAUGHTING SERVICES
J.A.M. DE WAAL. 12 BRIGHTWELL STREET PAPAKURA 2110.
NEW ZEALAND. PHONE: 0064 09 2988815. MOB: 0211791000
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PROJECTION
JDWDS
DATE DECEMBER-2017
SHEET: 04 OF 04

MODEL SCALE: 1:1
DWG SCALE: 1:1 @A3 OR AS SHOWN

A3 No: SCRAPBOX-04